

Panel Data Replication Study

Mertzanis (2024): “Central Bank Policies and Green Bond Issuance on a
Global Scale”

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Abstract

This report partially replicates Mertzanis (2024), who studies the effect of central bank green policies on green bond issuance across 53 countries (1991–2021). We obtain the Mendeley dataset (76 countries total; 23 excluded for insufficient consecutive observations) and apply five panel estimators: Pooled OLS, Year FE, Entity FE, Two-Way FE (TWFE), and First Differences (FD), plus IV and dynamic panel extensions. The three main central bank policy variables (**cbmnd**, **cbenf**, **cbsys**) are *time-invariant* – 0% within-country variance – and are mechanically dropped by entity-FE estimators. In pooled OLS, none is statistically significant. Only **cbenf** is weakly significant (-0.177 , $p=0.02$) in the Year FE specification. The most robust finding is the financial integration index (**fie**): $+22.31^{***}$ in entity FE, $+7.54^{**}$ in TWFE, and $+27.45^{***}$ in First Differences. All IV first-stage F-statistics are below 10 (weak instruments). The autocorrelation of Δbndgrn ($= -0.513$) triggers Anderson-Hsiao; however, $\Delta X_{\text{CB}} = 0$ throughout (CB variables are time-invariant), making the dynamic extension degenerate for CB variables.

1 Introduction

1.1 Paper and research question

Mertzanis (2024) asks: do central bank green policies stimulate green bond issuance? Using a country-level panel of 46 countries over 2007–2021, the paper finds that CB mandates and enforcement policies positively predict green bond volumes after controlling for financial development, governance, and macroeconomic conditions.

1.2 Summary of key results

The critical methodological constraint is that the three CB policy variables are time-invariant: constant within each country across all years. Entity-FE estimators drop them by construction. Table 1 summarises the main coefficients from our replication.

Table 1: Key findings from replication

Estimator	Variable	Coeff.	Sig.
Pooled OLS	cbmnd	−0.043	
Pooled OLS	cbenf	−0.096	
Year FE	cbenf	−0.177	**
Between OLS	fie	−0.960	
Entity FE	fie	+22.31	***
Mundlak RE	fie	+21.45	***
TWFE	fie	+7.540	**
First Differences	fie	+27.45	***
IV (cbmnd)	cbmnd	−0.516	*
IV (cbenf)	cbenf	−0.823	**

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Cluster-robust SE by country.

2 Database

2.1 Data source and variables

Data come from Mertzanis (2024), deposited on Mendeley Data (doi:10.17632/xb2m4m8h24.1). The raw file contains 76 countries with up to 16 annual observations. The regression sample (complete controls) comprises $N=39$ countries, $NT=206$ observations.

Table 2: Variable definitions

Name	Description	Source
bndgrn	Log total green bond issuance (USD bn)	Climate Bonds Initiative
cbmnd	CB green mandate index (0–8)	Mertzanis (2024)
cbenf	CB enforcement policy index (0–9)	Mertzanis (2024)
cbsys	CB systemic risk index (0–3)	Mertzanis (2024)
fie	IMF financial integration index	IMF
rating	Moody’s sovereign rating (numeric)	Moody’s
gdpg	GDP growth rate (%)	World Bank
infl	CPI inflation (%)	World Bank
dum9	EU membership dummy (0/1)	World Bank
dum12	OECD membership dummy (0/1)	World Bank

2.2 Summary statistics

Table 3: Descriptive statistics (regression sample, N=206)

Variable	N	Mean	SD	Min	Median	Max
bndgrn	206	0.109	1.853	−4.914	0.220	3.723
cbmnd	206	4.733	1.773	0.000	4.000	8.000
cbenf	206	5.126	1.833	0.000	6.000	6.000
cbsys	206	1.029	0.997	0.000	1.000	3.000
infl	206	2.313	4.904	−28.620	1.650	32.300
rating	206	3.733	0.475	2.000	4.000	4.000
fie	206	0.793	0.107	0.472	0.824	0.939
gdpg	206	1.140	3.748	−14.868	1.963	8.516
dum9	206	0.384	0.487	0.000	0.000	1.000
dum12	206	0.743	0.438	0.000	1.000	1.000

3 Sample Selection

3.1 Consecutive-observation rule

We retain only country-year observations where the country has at least 3 consecutive observations for `bndgrn`. Of 76 countries, 23 are excluded (e.g. Bangladesh, Nigeria, Romania) – smaller or less advanced markets with only sporadic green bond activity.

Table 4: Sample selection

Step	Countries	Obs
Raw dataset	76	—
After consecutive-obs rule (≥ 3)	53	322
Regression sample (complete controls)	39	206

3.2 Distribution of time observations

The cleaned panel is strongly unbalanced. Most countries enter after 2013 as the green bond market grew. Twenty of 53 countries have only 3–4 annual observations; 8 countries have 7–11 observations.

Table 5: Distribution of T per country

T	Countries	Obs
1	5	5
2	2	4
3	7	21
4	13	52
5	8	40
6	7	42
7	8	56
8–11	3	29
Total	53	322

Observations per year rise from N=1 (1991–1994) to N=51 (2021), reflecting rapid green bond market growth post-2015.

4 Variable Classification

4.1 Variance decomposition

Table 6: Variable classification by within-variance share

Category	Variables	Count
Time-invariant (0% within)	cbmnd, cbenf, cbsys, dum9, dum12	5
Mostly between (<10%)	rating (4.2%), fie (7.7%)	2
Mostly within (>50%)	bndgrn (59.4%), gdpg (62.0%), infl (75.5%)	3

Table 7: Variance decomposition (sorted by within %)

Variable	N	NT	Var Between	Var Within	Within %
cbmnd	53	283	3.302	0.000	0.0%
cbenf	53	283	—	0.000	0.0%
cbsys	53	283	—	0.000	0.0%
dum9	53	322	0.215	0.000	0.0%
dum12	53	322	0.254	0.000	0.0%
rating	53	316	0.366	0.011	4.2%
fie	53	289	0.010	0.001	7.7%
bndgrn	53	322	1.570	2.149	59.4%
gdpg	53	302	23.34	17.50	62.0%
infl	53	291	6.554	14.35	75.5%

Critical implication: cbmnd, cbenf, cbsys, dum9, and dum12 have exactly 0% within-country variance. Entity-FE estimators drop them by construction. Only Pooled OLS,

Year FE, Between OLS, and Mundlak RE can identify their coefficients.

5 Descriptive Statistics

5.1 Univariate statistics across transformations

Table 8: Descriptive statistics: **bndgrn** across transformations

Transform	N	Mean	Std	p25	Median	p75
Between	322	0.120	1.213	−0.622	0.155	0.938
One-Way Within	322	0.000	1.466	−0.885	0.196	0.957
First Diff	269	0.473	1.411	−0.271	0.403	1.157

Note: The balanced-TWFE subsample contains only 15 observations (all variables collapse to near-zero variance), so TWFE descriptive statistics are degenerate. Regression TWFE uses the Wansbeek-Kapteijn (1989) unbalanced method instead.

5.2 Correlation matrices (key entries)

Between: $r(\text{bndgrn}, \text{fie}) = +0.207$. CB dummies: *cbmnd* −0.07, *cbenf* −0.13, *cbsys* +0.02 (all insignificant).

One-way within: $r(\widetilde{\text{bndgrn}}, \widetilde{\text{fie}}) = +0.387$.

First differences: $r(\Delta \text{bndgrn}, \Delta \text{fie}) = +0.048$.

5.3 Country heterogeneity

Because CB variables are time-invariant, $\Delta X_{\text{CB}} = 0$ for all observations and country-level correlations $r(\Delta Y, \Delta X_{\text{CB}})$ are undefined. Heterogeneity analysis focuses on **fie**. Countries such as Germany ($r = 0.71$ in TWFE) and Norway ($r = 0.65$) show strong positive within-country correlations. A small number (Peru, Turkey) show negative correlations, likely reflecting noise in short series (T=3–4).

6 Panel Estimation Results

6.1 Main regression table (replication of Paper Table 4)

Table 9: Panel regression results: replication of Paper Table 4 (dep. var.: `bndgrn`)

	(1) Pooled	(2) +Year FE	(3) +Entity FE	(4) TWFE
<code>cbmnd</code>	−0.043 (0.127)	−0.087 (0.127)		
<code>cbenf</code>	−0.096 (0.071)	−0.177** (0.072)		
<code>cbsys</code>	−0.024 (0.235)	−0.203 (0.251)		
<code>infl</code>	−0.033 (0.039)	+0.011 (0.019)	−0.045 (0.038)	+0.007 (0.013)
<code>rating</code>	+0.642 (0.412)	+1.093** (0.463)	+0.783*** (0.219)	+0.771* (0.428)
<code>fie</code>	+1.829 (1.679)	−0.207 (1.943)	+22.31*** (4.447)	+7.540** (3.630)
<code>gdp_g</code>	−0.046 (0.031)	+0.022 (0.059)	−0.016 (0.023)	+0.056 (0.035)
<code>dum9</code>	+0.488 (0.532)	+0.643 (0.578)		
<code>dum12</code>	−0.135 (0.574)	+0.274 (0.572)		
Entity FE	No	No	Yes	Yes
Time FE	No	Yes	No	Yes
N (obs)	206	206	224	224
N (countries)	39	39	42	42
R ²	0.101	0.415	0.248	0.044

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Cluster-robust SE in parentheses. Blank cells = dropped due to collinearity with entity dummies (correct behaviour).

6.2 Homework estimator comparison

Table 10: Five-estimator comparison (homework specification)

Estimator	N obs	N ctry	fie coef	SE	Sig
Between OLS	39	39	−0.960	1.628	
One-Way FE	206	39	+21.455	4.588	***
Mundlak RE	206	39	+21.455	3.894	***
TWFE	206	39	+7.880	3.520	**
First Differences	144	39	+27.446	5.255	***

CB variables are NaN in FE/TWFE/FD. In Between and Mundlak RE: all negative, none significant ($p > 0.25$).

6.3 Notes on CB policy variables

In Pooled OLS all three CB variables are insignificant ($p > 0.10$). In Year FE only **cbenf** achieves weak significance (-0.177^{**} , $SE=0.072$). Negative signs contrast with Mertzanis (2024).

7 Robustness Checks (Paper Table 5)

Table 11: Robustness checks

	(2) Corp+EntFE	(3) Developed	(5) Ex-China	(6) Non-linear
cbmnd		-0.154	+0.103	-1.206***
cbenf		-0.196**	-0.192**	-0.495
cbsys		-0.214	-0.035	+0.489
fie	+22.65***	-0.000	+0.018	+0.292
rating	+0.793***	+1.283***	+0.665	+0.975**
cbmnd ²				+0.127***
N obs	224	196	199	206
R ²	0.245	0.443	0.455	0.487

The non-linear spec reveals a U-shaped **cbmnd** relationship: linear term -1.206^{***} and squared term $+0.127^{***}$, suggesting moderate mandates suppress issuance but very broad mandates may eventually support it.

8 Additional Controls (Paper Table 7)

Table 12: Additional controls (year FE + pooled OLS)

	(1) +Risk premia	(2) +Energy/Env	(3) +Geopolitical
cbmnd	−0.107	−0.087	−0.078
cbenf	−0.144**	−0.161*	−0.157*
cbsys	−0.219	−0.220	−0.264
fie	−1.946	+0.830	−0.612
rating	−0.399	+0.988*	+1.200**
riskeqt	−20.54		
riskctr	−72.28		
enrgoil		−0.029	
envtax		−0.371	
gprlog			+0.856**
wui			−0.732
N obs	206	181	196
R ²	0.450	0.489	0.416

cbenf remains significant at 10% in all three columns. The geopolitical risk index (**gprlog**: +0.856**, SE=0.322) is the only additional control achieving 5% significance.

9 Hausman Test and Estimator Choice

The Hausman test comparing One-Way FE against standard RE favours FE (rejection at 5%). Mundlak RE yields **fie** = 21.455 in both specifications, confirming internal consistency. FE is the preferred estimator for causal inference on time-varying regressors.

10 Time-Invariant Variables: IV and Interactions

10.1 IV / 2SLS on country means (Paper Table 6)

Table 13: IV / 2SLS results (country-means, cultural instruments)

Spec	N	1st-F	cbmnd	cbenf
(1) cbmnd $\sim$ hierarchy+purity	34	5.10	−0.516*	−0.140
(2) cbenf $\sim$ hierarchy+openness	27	4.45	+0.372*	−0.823**
(3) cbsys $\sim$ hierarchy+openness	27	1.75	+0.223	−0.152

Weak instruments: all F-stats below 10. IV estimates should be interpreted with caution.

10.2 Interaction models (Paper Table 9)

Table 14: cbenf $\times$ country characteristic

Interaction	N	Interact. coef	p-value
cbenf $\times$ prodstr	205	−0.088	0.562
cbenf $\times$ bigtech	152	+0.0017	0.086*
cbenf $\times$ ngain	165	−0.029	0.352

Only the **bigtech** interaction is marginally significant: the negative **cbenf** effect is attenuated in countries with larger tech sectors.

11 Dynamic Panel Estimation (Anderson-Hsiao)

The autocorrelation of Δbndgrn is $r(\Delta Y_t, \Delta Y_{t-1}) = -0.513$ ($p < 0.001$), exceeding the 0.10 trigger threshold. Anderson-Hsiao (1981) is applied, using $Y_{i,t-2}$ and $X_{i,t-2}$ as instruments.

Important caveat: $\Delta X_{\text{CB}} = 0$ for all observations. The dynamic regressor dX is degenerate; impulse responses and the long-run coefficient are undefined.

Table 15: Dynamic panel: key parameters

Parameter	OLS	SE	IV (AH)	SE
$\beta_y (\Delta Y_{t-1})$	−0.381	0.089	+0.039	0.200
Δfie	+11.19	5.320	+6.547	5.035
Δrating	−2.141	0.246	−3.172	0.515
Δgdpg	+0.072	0.037	+0.123	0.049
N obs / N countries	131 / 34		131 / 34	
1st-stage $R^2 (\Delta Y_{t-1})$	—		0.375	

The OLS estimate $\beta_y = -0.381^{***}$ indicates strong mean-reversion. The IV estimate (+0.039, $p > 0.10$) is not significant. The **fie** effect remains positive in both specs but loses significance under IV.

12 Discussion

12.1 Comparison with Mertzanis (2024)

CB policy variables: Time-invariance prevents within-country identification. Between-country signs are negative (opposite to Mertzanis). Only **cbenf** achieves weak significance (−0.177**) in Year FE; IV confirms negative point estimates but faces weak-instrument concerns.

Financial integration: Entity FE (+22.31***) shows that within a country, higher `fie` strongly predicts higher green bond issuance. Consistent across TWFE (+7.54**) and FD (+27.45***). Large magnitude partly reflects the small within-variance of `fie` (7.7%).

Sovereign rating: Positive across entity-FE specifications (+0.783***). Better credit quality enables lower-cost green bond issuance.

12.2 Limitations

Time-invariant CB variables cannot be identified by FE. Short average T limits TWFE precision. All IV first-stage F-stats below 10 (weak instruments). Country heterogeneity is large; pooled coefficients may not represent any individual country. Dynamic extension is degenerate for CB variables.

13 References

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A Appendix A: Replication Checklist

#	Section	Task	Status
1	Data	Download Mendeley dataset (76 countries)	Done
2	Data	Load, inspect, rename variables	Done
3	Clean	Consecutive-obs rule (≥ 3); N=53, NT=322	Done

#	Section	Task	Status
4	Clean	Sample selection table	Done
5	Trans.	Between transformation	Done
6	Trans.	Within transformation	Done
7	Trans.	First differences (NaN at individual breaks)	Done
8	Trans.	TWFE via Wansbeek-Kaptein (unbalanced)	Done
9	Var.	Variance decomposition; 5 time-invariant vars found	Done
10	Desc.	Univariate stats (all 4 transforms)	Done
11	Desc.	Correlation matrices	Done
12	Desc.	Country heterogeneity tables (fie; CB=0 by design)	Done
13	Reg.	Pooled OLS	Done
14	Reg.	Year FE	Done
15	Reg.	Entity FE	Done
16	Reg.	TWFE	Done
17	Reg.	Between OLS (homework)	Done
18	Reg.	Mundlak RE (homework)	Done
19	Reg.	First Differences (homework)	Done
20	Rob.	Paper Table 5 robustness	Done
21	Rob.	Paper Table 7 additional controls	Done
22	IV	Paper Table 6 IV/2SLS; all $F < 10$ (weak)	Done
23	Intx.	Paper Table 9 interactions	Done
24	Dyn.	AR(1) = -0.513 triggers Anderson-Hsiao	Done
25	Dyn.	Anderson-Hsiao IV; dX degenerate (CB time-invariant)	Done
26	Hausman	FE vs RE: FE preferred	Done

B Appendix B: Panel Data To-Do List

1. **Identify variables.** Record Y, X, controls; check time-invariance before choosing estimators.
2. **Check within variance.** Variables with 0% within variance are dropped by entity

FE/TWFE. Plan IV or between strategies.

3. **Apply sample selection.** Require ≥ 3 consecutive obs of Y . Document excluded units and selection bias.
4. **Describe panel structure.** Attrition tables: N by date, T distribution, proportion with holes.
5. **Compute all four transformations.** Between, one-way within, FD (NaN at breaks), TWFE (Wansbeek-Kapteijn for unbalanced).
6. **Run five standard estimators.** Between OLS, Entity FE, Mundlak RE, TWFE, First Differences. Cluster SEs by entity.
7. **Run Hausman test.** Confirm Mundlak RE matches FE on time-varying regressors.
8. **Check AR(1) of ΔY .** If $|AR(1)| > 0.10$, use Anderson-Hsiao. Check first-stage $F > 10$.
9. **Handle time-invariant X .** Between/Pooled OLS for point estimates; IV with cross-sectional instruments (flag if $F < 10$).
10. **Report actual numbers only.** Never substitute fabricated or illustrative values for script output.